

Extrauterine Pregnancy 06. Ultrasound Pattern Recognition With Consequences

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 6. Ultrasound Pattern Recognition With Consequences.

Ultrasound is the central investigation in ectopic pregnancy because ectopic pregnancy is a question of implantation site. But ultrasound does not produce one kind of answer. Some findings prove the diagnosis. Some make it highly probable. Some only raise suspicion or show danger.

The resident must learn the diagnostic strength of each finding.

NICE gives the cleanest framework. A tubal ectopic pregnancy is diagnosed when an adnexal mass moves separately from the ovary and contains a gestational sac with a yolk sac or fetal pole, with or without fetal heartbeat. That is anatomy of pregnancy outside the normal location. When that anatomy is seen, hCG uncertainty should not obscure the diagnosis.

This is the strongest ultrasound category because it shows two things at once: pregnancy anatomy and wrong location. A yolk sac or fetal pole proves that the structure is not just a clot, corpus luteum, or nonspecific adnexal mass. Movement separate from the ovary proves that it does not belong to the ovary. The diagnosis is strong because the image has answered both identity and location.

Other findings have lower certainty. A complex inhomogeneous adnexal mass, an adnexal mass with an empty gestational sac, a tubal ring or bagel sign, and the blob sign can be highly suspicious, especially when the mass is separate from the ovary. Nature Reviews describes the blob sign as common among sonographically diagnosed ectopic pregnancies and notes its high positive predictive value. But a mass is still not the same as a yolk sac or embryo outside the uterus. It must be interpreted with symptoms, hCG, ovarian relationship, and follow-up safety.

Some findings are contextual rather than diagnostic. An empty uterus, a fluid collection in the uterine cavity, a pseudosac, or free peritoneal fluid may belong to the ectopic picture, but NICE emphasizes that these features are not reliable enough on

their own to diagnose ectopic pregnancy. They require integration with other intrauterine and adnexal features, the clinical presentation, and serum hCG.

This distinction should be spoken in the report and in the clinical discussion. "A definite ectopic pregnancy is seen" is different from "there is a high-probability adnexal mass" and different again from "there are possible supporting features." If the wording does not carry diagnostic strength, the treatment decision can become either too aggressive or too slow.

The phrase "separate from the ovary" deserves special attention.

A corpus luteum is common in early pregnancy and may look vascular, cystic, or ring-like. A tubal ectopic should move separately from the ovary when gentle probe pressure is applied. An ovarian ectopic, by contrast, is within ovarian parenchyma and cannot be separated from the ovary. ESHRE warns that small ovarian pregnancies can resemble a corpus luteum, and color Doppler may help distinguish corpus luteum from peri-trophoblastic flow.

So the scan is not simply asking, "Is there a mass?" It is asking what the mass belongs to. Identify the ovary. Identify the corpus luteum when possible. Decide whether the suspicious structure is ovarian, paraovarian, tubal, adherent, or separate. If the patient is stable and the relationship is unclear, repeat scanning or expert review is better than converting uncertainty into a confident label.

That repetition matters because many false decisions start with the word "mass." A mass can be pregnancy tissue, a corpus luteum, blood clot, endometrioma, hydrosalpinx, dermoid, fibroid, or inflammatory disease. The next question is never only size. The next question is belonging: does it belong to the ovary, to the tube, to the uterine wall, to the scar, or to the peritoneal cavity? Ultrasound becomes safer when the report describes relationships rather than only objects.

Farren recommends a stepwise ultrasound approach. First assess uterine shape and anomalies. Then assess the relationship between trophoblast and the uterine cavity. For uterine and interstitial ectopics, the key question is not merely whether something is "in the uterus," but whether the gestation communicates appropriately with the endometrial cavity and whether it is surrounded by normal tissue. Three-dimensional ultrasound may help when a laterally located pregnancy could be confused with interstitial ectopic pregnancy.

This is the same discipline introduced earlier in the course: a uterine address is not enough. The scan must define the implantation bed.

Interstitial pregnancy illustrates the problem. ESHRE describes the critical diagnostic feature as visualization of a thin intramural or interstitial segment of fallopian tube adjoining the medial aspect of the gestational sac and the lateral aspect of the uterine cavity, often called the interstitial line sign, with the sac at least partly enveloped by myometrium. Farren shows how a laterally located normally sited pregnancy may initially raise concern and then declare its connection to the cavity on repeat imaging.

In a stable patient with uncertainty, time and expert repeat imaging can prevent false-positive diagnosis.

Free fluid is a different ultrasound question. It does not necessarily locate the pregnancy. It tells you about bleeding risk.

ESHRE recommends describing hemoperitoneum semi-quantitatively. Mild means echogenic fluid in the pouch of Douglas. Moderate means visible blood clots. Severe means blood clots and echogenic fluid in both the pouch of Douglas and the uterovesical space. Blood in Morrison pouch, the hepatorenal space, suggests severe intra-abdominal bleeding.

Make the fluid grading audible. Mild is echogenic fluid only in the pouch of Douglas. Moderate is no longer just fluid; there are clots. Severe means blood and clots extend beyond the posterior pelvis into more than one pelvic space, and Morrison pouch blood means the upper abdomen has joined the story. The location of fluid tells you how much bleeding may have escaped the pelvis. This is why free fluid is not an incidental comment in an ectopic scan.

That grading matters because a scan can fail to prove the exact pregnancy location and still show enough blood to make the patient unsafe. Conversely, a scan can show a small ectopic pregnancy with no free fluid in a stable patient, giving time to discuss medical or expectant options. Location tells you what the pregnancy is. Fluid, clot, vital signs, and symptoms tell you how much time the patient has.

Measurement also affects management. ESHRE recommends documenting morphology: whether there is a gestational sac, yolk sac, fetal pole, cardiac activity, or only solid swelling. It also recommends measuring the ectopic pregnancy, trophoblastic mass, gestational sac if present, and hematosalpinx when present. Farren explains that measuring only the small coelomic cavity can understate the clinically relevant mass when most of what is seen is trophoblast or hematosalpinx.

This is not a radiology detail. If the measured size affects eligibility for expectant or medical management, the resident must know what was measured.

Operator dependence must be treated honestly. Farren notes that expert units may identify a high proportion of extrauterine ectopics on the initial scan, but rare ectopics such as intramural or abdominal pregnancies remain difficult because most clinicians have limited experience with them. ESHRE explicitly calls for collaboration and expert-center advice when complex or rare ectopic pregnancy is suspected. The safe response to diagnostic uncertainty is not pride. It is escalation of expertise.

A practical ultrasound report for suspected ectopic pregnancy should answer several questions.

Is there a normally sited pregnancy in the endometrial cavity? If not, is there definite ectopic pregnancy anatomy: gestational sac, yolk sac, embryo, or cardiac activity outside the normal site? If not, is there a suspicious adnexal mass, and does it move

separately from the ovary? What is the corpus luteum? Is there free fluid, is it echogenic, are there clots, and does it extend beyond the pelvis? Could the location be interstitial, scar, cervical, intramural, ovarian, abdominal, or rudimentary horn? Are the images good enough, or does the patient need expert review?

If the listener remembers only one ultrasound habit, it should be this sequence: prove normal location if possible; if not, look for definite ectopic anatomy; if not definite, describe probability and danger. Probability comes from the mass and its relationships. Danger comes from bleeding, symptoms, and stability. A report that separates certainty from danger helps the clinician choose follow-up, methotrexate, surgery, or emergency escalation without pretending that all ultrasound findings are equal.

Treatment should follow the strength of the evidence. Definite anatomy can justify definitive management when the clinical setting fits. Probable findings require integration. Possible findings require follow-up or escalation, not pretending. Hemoperitoneum and physiology can convert diagnostic uncertainty into emergency action.

Ultrasound is not only pattern recognition. It is pattern recognition with consequences: the report should tell the clinician what has been proven, what is probable, what remains uncertain, and what danger is already visible.